

TriCognia Ville

Educational Game (Android)

PROJECT TYPE Educational Game (Android)	PLATFORM Landscape Android game with web backend
AUDIENCE Grade 7 readers and classroom educators	PRIMARY FOCUS Gamified reading comprehension

Overview

A story-based gamified reading application that helps Grade 7 learners rebuild a faded village by completing reading and comprehension quests. As students master each task, buildings return from grayscale to full color, livening the town with characters, lights, and ambient activity. Built as a landscape Android app in Godot 4.6 with a Next.js 16 + Neon PostgreSQL backend and Groq-powered adaptive feedback.

Core Features

- 6 Quest Types: Multiple Choice, Drag-and-Drop, Read-Aloud, Fluency Drill, Tap-Target, Punctuation Read
- Grayscale-to-color village restoration via custom canvas-item shader
- AI-powered adaptive feedback and hint scaffolding via Groq
- Speech-recognized read-aloud and fluency drills with mic-aware audio ducking
- Story-driven narrative arc with reflection prompts and unlock cutscenes
- JOSE-signed JWT auth with bcrypt and per-learner data isolation
- Cloud sync over Drizzle ORM + Neon PostgreSQL with offline backoff
- Printable completion certificates generated with pdf-lib

Experience and Design

- Learners restore a faded village by completing six types of reading quests.
- A story-driven sequence connects assessment, tutorials, practice, missions, and reflection.
- Read-aloud and fluency activities combine speech recognition with targeted feedback.
- Badges, restoration states, cutscenes, and certificates make progress tangible.

Architecture and Data

- Godot and GDScript power the landscape game, quest system, shaders, dialogue, and audio.
- A Next.js backend exposes authenticated progress and content endpoints.
- Drizzle ORM stores learner records in Neon PostgreSQL.
- Groq services generate adaptive hints while local saves provide offline resilience.

Security and Privacy

- Bcrypt credentials and JOSE-signed JWTs protect learner accounts.
- Zod validation protects backend requests and progress submissions.
- Learner attempts, progress, and certificates remain isolated per account.
- Local-first saves synchronize through authenticated HTTPS requests.

Technology Stack

Godot 4.6	Application framework and UI
GDScript	Application language
Next.js 16	Application framework and UI
TypeScript	Application language
PostgreSQL (Neon)	Authentication, database, or cloud data
Drizzle ORM	Database access and schema management
Groq SDK	AI and speech processing
Tailwind CSS	Interface styling and design system

Performance and Reliability

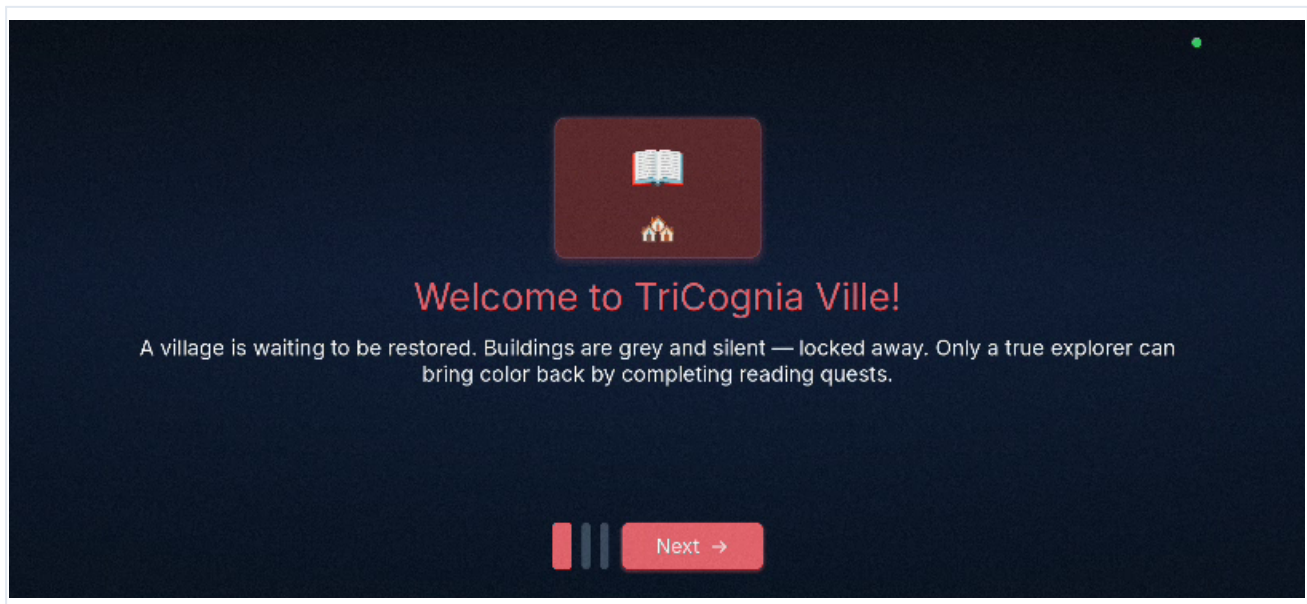
- The compatibility renderer supports a broad range of Android tablets.
- Per-sprite grayscale shaders avoid an expensive full-screen post-processing pass.
- Quest content and media load only when the related activity begins.
- Offline backoff prevents repeated failed synchronization attempts.

Platform Requirements

- Android device or tablet with landscape orientation support.
- Microphone permission for read-aloud and fluency activities.
- Internet connection for account sync and adaptive AI feedback.
- Audio output is recommended for dialogue, narration, and pronunciation tasks.

PAGE OVERVIEW

Onboarding

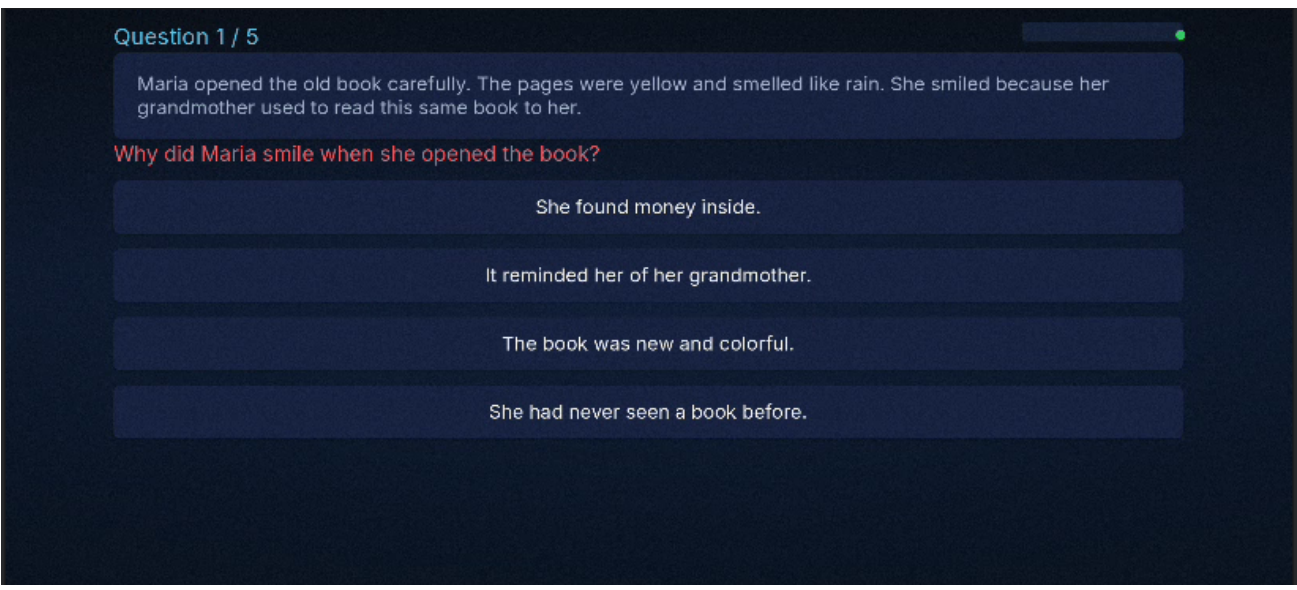


Onboarding

Interface view from TriCognia Ville.

PAGE OVERVIEW

Reading Assessment



Question 1 / 5

Maria opened the old book carefully. The pages were yellow and smelled like rain. She smiled because her grandmother used to read this same book to her.

Why did Maria smile when she opened the book?

She found money inside.

It reminded her of her grandmother.

The book was new and colorful.

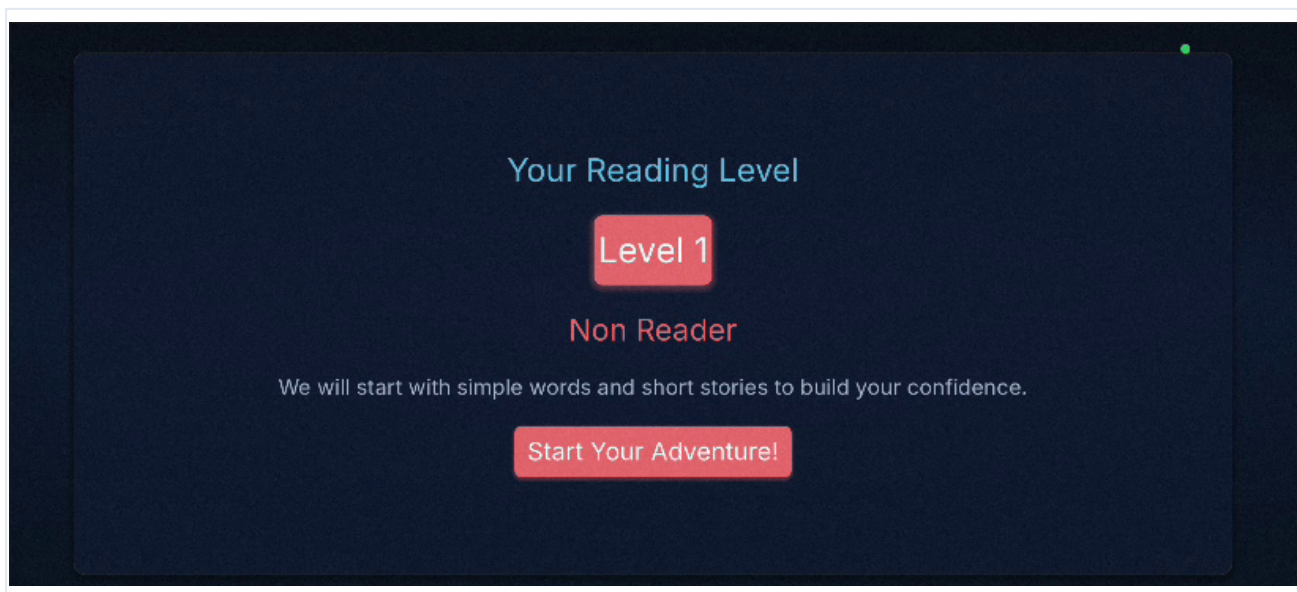
She had never seen a book before.

Reading Assessment

Interface view from TriCognia Ville.

PAGE OVERVIEW

Reading Level Reveal



Reading Level Reveal

Interface view from TriCognia Ville.

PAGE OVERVIEW

Village - Faded Start

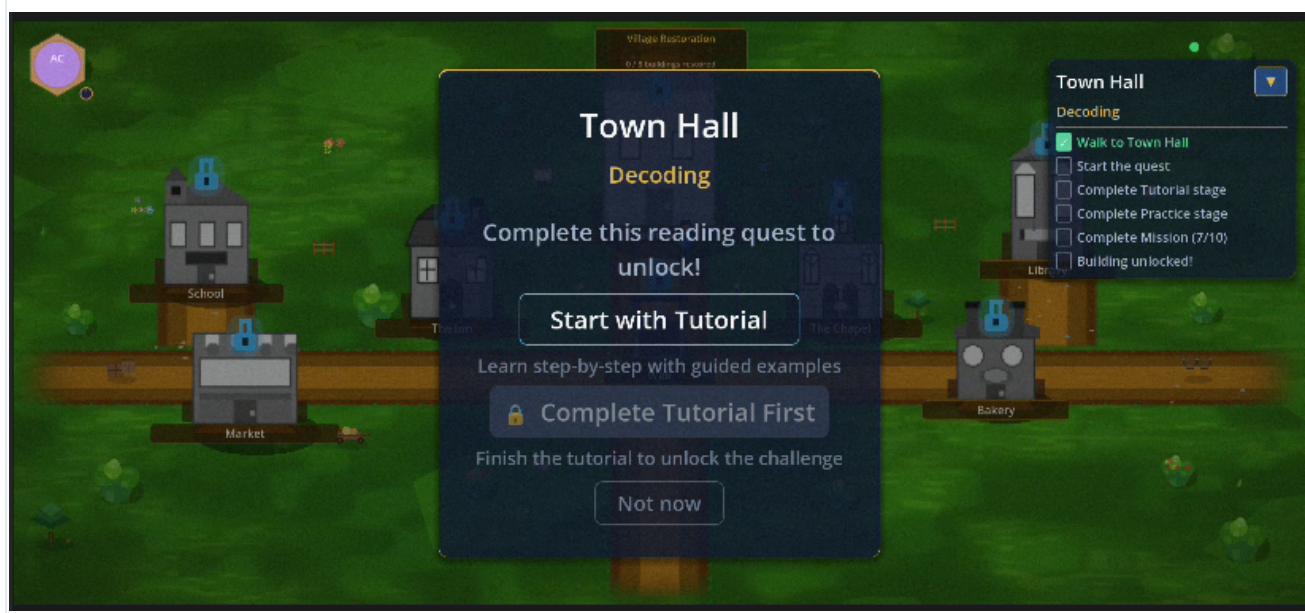


Village - Faded Start

Interface view from TriCognia Ville.

PAGE OVERVIEW

Tutorial / Practice / Mission Picker

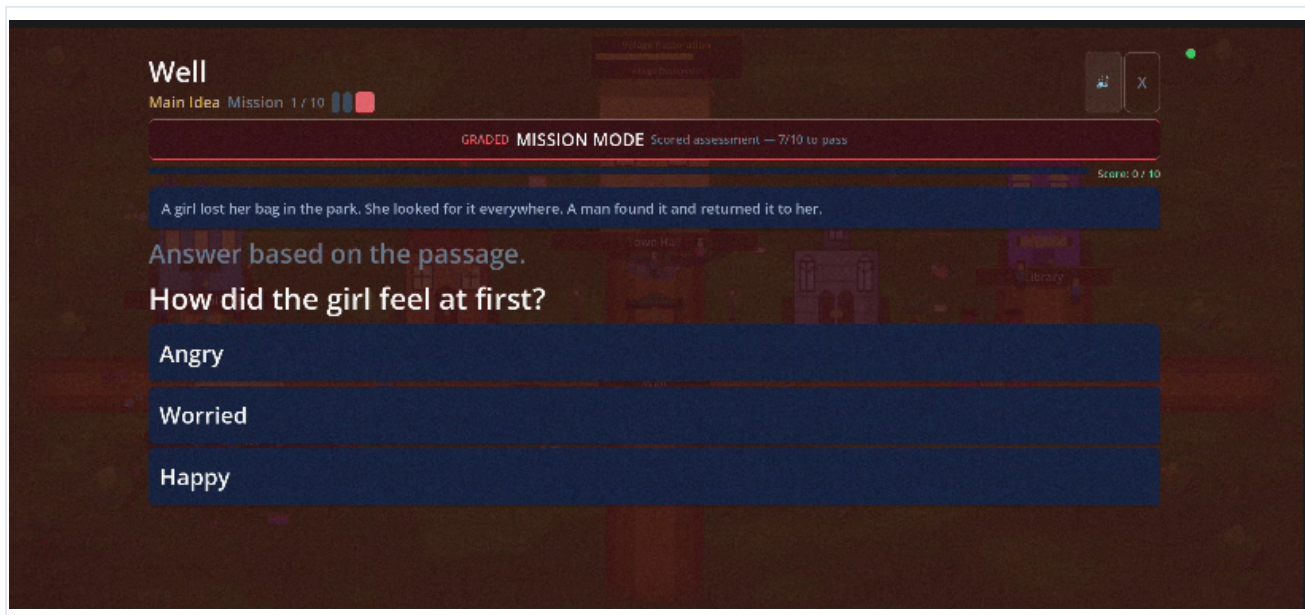


Tutorial / Practice / Mission Picker

Interface view from TriCognia Ville.

PAGE OVERVIEW

Mission Mode



Mission Mode

Interface view from TriCognia Ville.

PAGE OVERVIEW

Story Dialogue

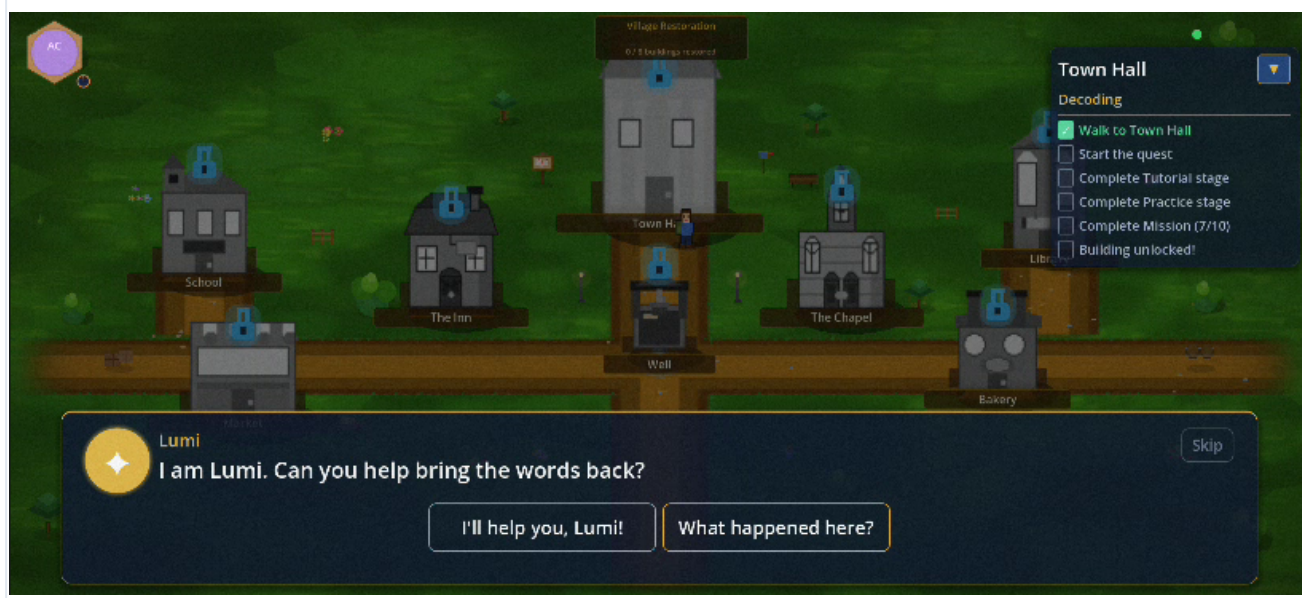


Story Dialogue

Interface view from TriCognia Ville.

PAGE OVERVIEW

Story Dialogue (cont.)

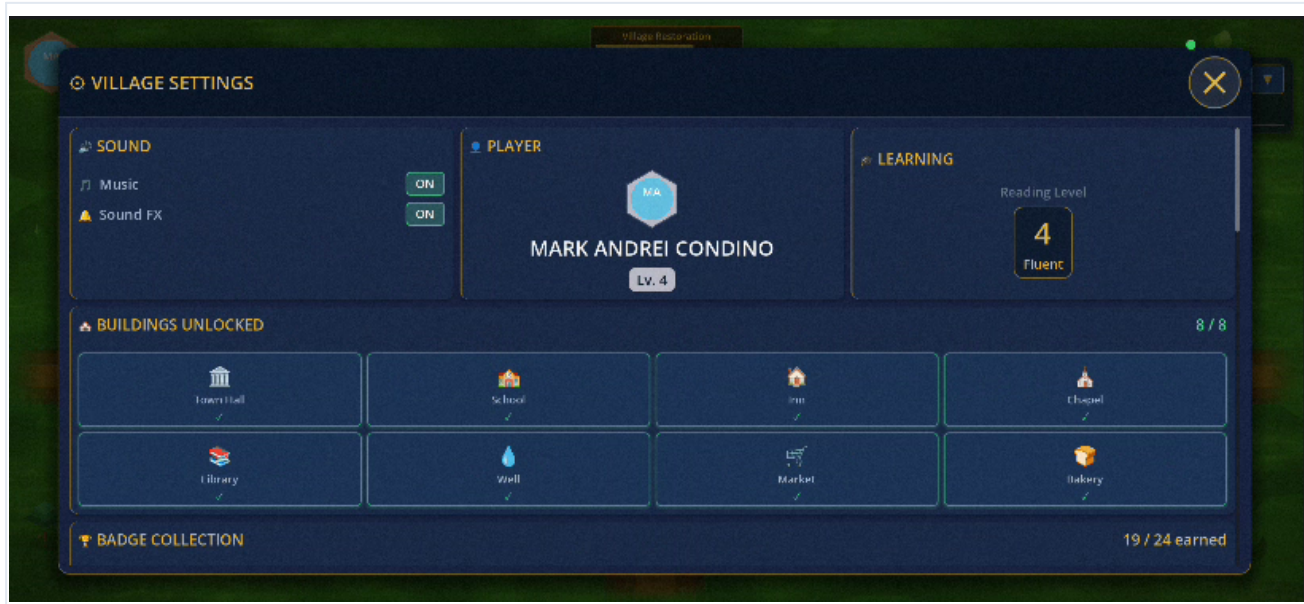


Story Dialogue (cont.)

Interface view from TriCognia Ville.

PAGE OVERVIEW

Player Profile & Badges

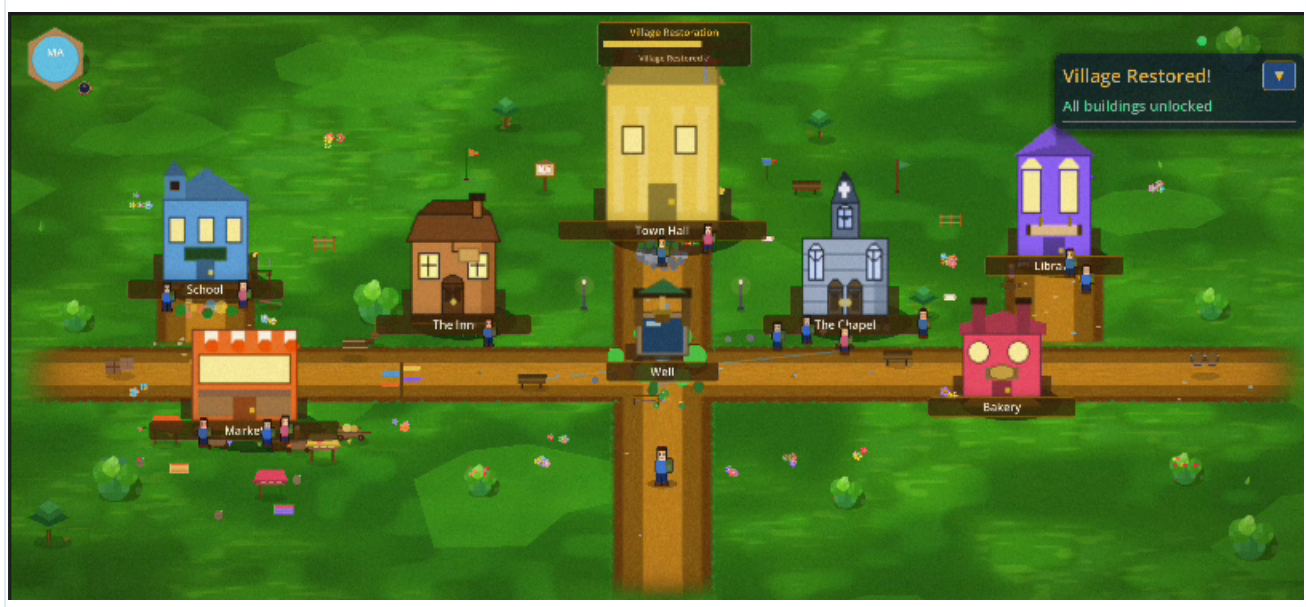


Player Profile & Badges

Interface view from TriCognia Ville.

PAGE OVERVIEW

Village - Restored



Village - Restored

Interface view from TriCognia Ville.